

The SCARE Framework: Creating Effective AI Prompts for Teaching and Learning

Jonathan R. Bacolod

BRACE YOURSELF

TRIVIA IS COMING....

Did you know?

APP

KEY IDEA

The Metric System (SI)

The **International System of Units (SI)** is the standard system of measurement used worldwide.

Metric units are related by powers of 10 — that's what makes converting so simple:



Larger → Smaller Unit

MULTIPLY by the power of 10

(move the decimal point RIGHT)



Smaller → Larger Unit

DIVIDE by the power of 10

(move the decimal point LEFT)

Example 1

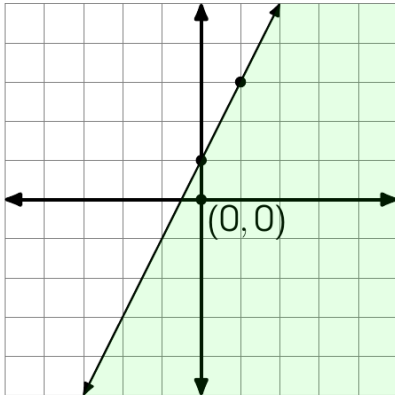
Step 4: If the inequality is satisfied, shade the region containing that point. If not, shade the other region.

Original inequality $y \leq 2x + 1$

Plane divider $y = 2x + 1$

$m = 2$

$b = 1$



Let $x = 0, y = 0$

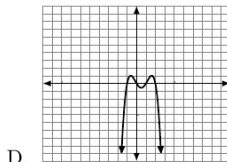
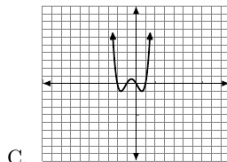
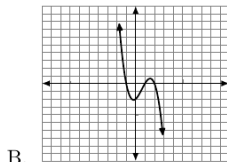
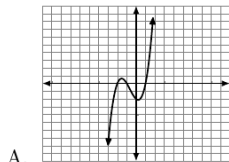
$y \leq 2x + 1$

$0 \leq 2(0) + 1$

$0 \leq 0 + 1$

$0 \leq 1$ ✓

16. Which of the following could be the graph of the polynomial function $y = -x^3 + 2x^2 + x - 2$?



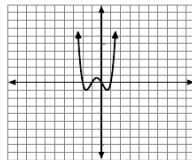
17. What are the end behaviors of the graph of $f(x) = -2x + x^3 + 3x^5 - 4$?

- A. rises to the left and falls to the right
B. falls to the left and rises to the right

- C. rises to both directions
D. falls to both directions

18. Which polynomial function in factored form represents the given graph?

- A. $y = (x + 2)(x + 1)(x - 1)$
B. $y = (x - 2)(x + 1)(x - 1)$
C. $y = x(x + 2)(x + 1)(x - 1)$
D. $y = x(x - 2)(x + 1)(x - 1)$



19. An arc with a measure equal to one-half the circumference of a circle is called:

- A. Intercepted arc B. Major arc C. Minor arc D. Semicircle

20. If $m\widehat{AC} = 40^\circ$ and $m\widehat{BD} = 80^\circ$, find $m\angle ATC$.



Mathematics 8

Module 6 – Linear Equations

Introduction: This module was designed and written with you in mind. It is specially made for you to gain a thorough understanding on several concepts on linear equations. Specifically, you will be able to write the linear equation $Ax + By = C$ in the form $y = mx + b$ and vice versa, graph a linear equation given (a) any two points; (b) the x- and y- intercepts; (c) the slope and the y-intercept; and (d) the slope and a point on the line, and describe the graph of a linear equation in terms of its intercepts and slope.

Lesson 1

Forms of Linear Equation



What To Know

This module is designed for grade 8 students like you, so that you will have a better understanding on the concepts of linear equations. Specifically, you are expected to:

1. write the linear equation $Ax + By = C$ in the form $y = mx + b$ and determine the values of m and b ;
2. write the linear equation $y = mx + b$ in the form $Ax + By = C$ and determine the

3. graph a linear equation given the slope and the y-intercept;
4. graph a linear equation given the slope and a point on the line.

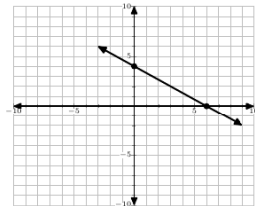


Pre-Assessment

Answer the following questions. Choose the letter that corresponds to the correct answer.

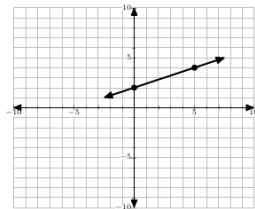
1. Which of the following information may be used to plot the graph at the right?

- A. $a = 4$ and $b = 6$
- B. $m = 4$ and $b = 6$
- C. $a = 6$ and $b = 4$
- D. $m = 6$ and $b = 4$



2. The graph at the right can be described by which of the following information?

- A. $a = 2$ and $m = \frac{2}{5}$
- B. $b = 2$ and $m = \frac{2}{5}$
- C. $a = 2$ and $b = \frac{2}{5}$
- D. $b = 2$ and $a = \frac{2}{5}$



**“Math Hugot”:
An Intervention for Improving
Attitudes toward Mathematics Among
Grade 10 Students of
Sauyo High School,
S.Y. 2019–2020**

Jonathan R. Bacolod, LPT

Researcher

Quiz 3.6: Proving the Congruence of Triangles

Multiple Choice: Choose the letter that corresponds to the correct answer. Write the answer in your answer sheet.

1. Which theorem states that if the legs of one right triangle are congruent to the legs of another right triangle, then the triangles are congruent?

A. HA Congruence Theorem

C. LA Congruence Theorem

B. HL Congruence Theorem

D. LL Congruence Theorem

2. "If two angles and a non-included side of one triangle are congruent to the corresponding two angles and a non-included side of another triangle, then the triangles are congruent." This is stated in:

A. AAS Congruence Theorem

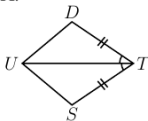
C. HL Congruence Theorem

B. LL Congruence Theorem

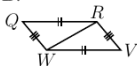
D. LA Congruence Theorem

3. Which of the following pairs of triangles are congruent and can be proved by ASA Postulate?

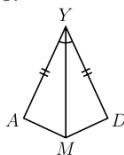
A.



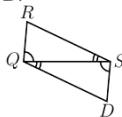
B.



C.



D.



4. Which of the following pairs of triangles are congruent and can be proved by HL Theorem?

Worksheet 1.5.3: Slope of a Line

A. True or False

Write True if the statement is true or False if it is false. One point each.

1. The slope of the line described by the equation $y = -3x$ is -3 .
2. The slope of the line described by the equation $y = 4x - 1$ is -1 .
3. The slope of the line described by the equation $y = \frac{3}{2}x + 2$ is 3 .
4. The slope of the line described by the equation $y = -\frac{5}{4}x - 3$ is $-\frac{5}{4}$.
5. The slope of the line described by the equation $5x + y = 3$ is 5 .

B. Finding the Slope Given Two Points

Find the slope of the line that passes through each pair of points. Choose the answer from the box. Some answers may be repeated. Write the letter only. One point each.

a. -4

e. $\frac{3}{4}$

b. -1

f. $\frac{3}{5}$

c. -10

g. $-\frac{3}{4}$

d. $\frac{15}{8}$

h. 1

6. $(4, 3)$ and $(2, 1)$

11. $(9, -7)$ and $(3, -1)$

7. $(-3, 2)$ and $(-5, 4)$

12. $(5, -4)$ and $(9, -7)$



	School	SAUYO HIGH SCHOOL	Grade Level	8
	Teacher	JONATHAN R. BACOLOD	Learning Area	Mathematics
	Teaching Date and Time	Nov 17, 2025 8 – Oman and Qatar 12:30 PM – 6:50 PM	Quarter	3rd

I. OBJECTIVES	
A. Content Standard	The learner demonstrates knowledge and understanding of linear equations in two variables and their graphs.
B. Performance Standard	The learner is able to graph linear equations in two variables. (NA)
C. Content Domain	Number and Algebra (NA)
D. Most Essential Learning Competencies/Objectives	find the equation of a line given the slope and a point
II. CONTENT	Finding the Equation of a Line Given the Slope and a Point
III. LEARNING RESOURCES	
A. References	
Teacher's Guide	Teacher-made Learning Worksheet
Learner's Materials	Activity Sheets
B. Other Learning Resources	Terpanel, Printed Materials, Activity Sheets

Republic of the Philippines
 Department of Education
 National Capital Region
 SCHOOLS DIVISION OFFICE QUEZON CITY
 SAUYO HIGH SCHOOL
 Pantabangan St., NIA Village, Sauyo, Quezon City

MATATAG K to 10 Curriculum Weekly Lesson Exemplar	School	SAUYO HIGH SCHOOL	Grade Level	8
	Name of Teacher	JONATHAN R. BACOLOD	Learning Area	Mathematics
	Teaching Dates and Time	December 8–12, 2025, 12:30 PM – 6:50 PM	Quarter	Quarter 3, Week 6

	DAY 1	DAY 2	DAY 3	DAY 4	DAY 5
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I. CURRICULUM CONTENT, STANDARDS, AND LESSON COMPETENCIES

A. Content Standards	The learners demonstrate knowledge and understanding of: 1. systems of linear equations in two variables
B. Performance Standards	By the end of the quarter, the learners are able to: 1. solve a system of linear equations graphically and algebraically
C. Learning Competencies	By the end of the lesson, the learners: 1. define and illustrate a system of linear equations in two variables 2. solve a system of linear equations (with integer solutions) by graphing 3. classify the types of systems of linear equations based on the number of solutions

	The learner should be able to: a. Recall the concepts	The learner should be able to: a. Outline and estimate	The learner should be able to: a. Solve and analyze the various	The learner should be able to: a. Cite and analyze the various components involved in classifying types of	The learner should be able to: a. Recall the concepts
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Inscribed Angles and Intercepted Arcs

Inscribed angle: an angle whose vertex lies on the circle and whose sides are chords of a circle

The measure of an inscribed angle is **half** the measure of its intercepted arc.

In a circle, if two inscribed angles intercept the same arc or congruent arcs, then the angles are congruent.

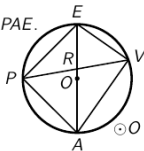
An angle inscribed in a semicircle is a right angle and therefore the measure is equal to 90° .

If a quadrilateral is inscribed in a circle, then its opposite angles are supplementary.

Practice Exercises

A. Refer to $\odot O$ to answer the following.

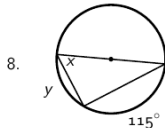
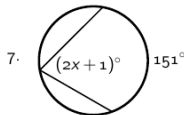
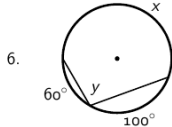
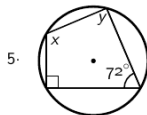
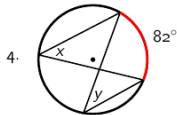
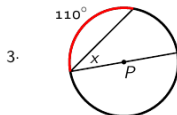
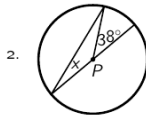
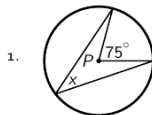
1. Name the angle that intercept $\widehat{AP}$.
2. Name the angles that intercept $\widehat{EV}$.
3. Name the arc that is intercepted by $\angle PAE$.
4. Name the arc that is intercepted by $\angle EVP$.
5. If $m\angle PEA = 48^\circ$, then $m\widehat{AP} = \underline{\hspace{2cm}}$
and $m\angle AVP = \underline{\hspace{2cm}}$.
6. $m\angle EPA = \underline{\hspace{2cm}}$
7. $m\angle EVP + m\angle PVA = \underline{\hspace{2cm}}$
8. If $m\angle VEP = 100^\circ$, then $m\angle PAV = \underline{\hspace{2cm}}$.



B. Given $\odot S, \overline{AR} \cong \overline{RO} \cong \overline{OS} \cong \overline{SA}, m\angle AMR = 3x + 20$ and

Problem Set

A. Use the given figures to find the value of x and y .



B. $\triangle GOA$ is inscribed in $\odot L$. If $m\angle OGA = 75^\circ$ and $m\widehat{AG} = 160^\circ$, find:

1. $m\widehat{OA}$

Grade 8 Math Handouts



3RD QUARTER

Lesson 3.1: Solving Linear Equations in One Variable

 Share Link

Lesson 3.1: Solving Linear Equations in One Variable

TeXMac⁴ Pro: Math Teaching Beautified

Linear equation: an equation where the variable is raised to the power of one, and the equation represents a straight line when graphed.

VOLUME OF A CYLINDER: FROM CIRCLE AREA TO 3D SPACE

Discovering $V = \pi r^2 h$ through Inductive Stacking & Pattern Recognition

1. What is Inductive Reasoning?

Inductive Reasoning means observing specific patterns and building up to a general formula!

Concept Bridge: Flat 2D Circle (Coin/Pancake) → Stacking h layers → 3D Solid Cylinder!

2. Graphic Organizer: The Pancake Stack

A circular coin/pancake has a base radius r and base area $A = \pi r^2$.

Inductive Stacking Process



💡 **GRADE 7 MEMORY HOOK (MNEMONIC)**

"PIES ARE SQUARED HIGH!" 🥧

Picture a tall stack of delicious circular pies piled up **HIGH!**

PIE (π) • **ARE** (r) • **SQUARED** (2) • **HIGH** (h) → $V = \pi r^2 h$

THE CYLINDER VOLUME FORMULA

$$V = \text{Base Area} \times \text{Height} = \pi r^2 h$$

Base Area (B) = Circle Area = πr^2



4. Worked Example: Soda Can Calculation

Problem: A soda can has a base radius $r = 3 \text{ cm}$ and height $h = 10$

Parallelism and Perpendicularity

Word List

A	T	T	F	C	J	B	A	P	S	R	Y	P	P	I	F	T	A	F	B
G	T	B	X	O	X	B	C	F	G	A	Y	E	L	S	B	V	L	C	H
I	M	A	T	N	G	D	J	K	L	N	Y	R	A	D	K	V	T	G	A
N	W	L	M	S	U	I	X	Y	O	O	M	P	M	U	N	X	E	E	N
T	K	T	P	E	N	S	E	D	U	F	Y	E	S	G	W	G	R	J	G
E	I	E	V	C	J	T	M	P	G	G	T	N	A	S	Z	Z	N	E	L
R	U	R	I	U	T	T	V	A	C	T	T	D	F	I	B	X	A	C	E
S	H	N	B	T	A	R	H	R	M	I	K	I	O	C	P	I	T	O	R
E	A	A	M	I	F	A	L	A	Y	C	S	C	C	T	B	Y	E	R	E
C	U	T	A	V	N	N	I	L	E	R	K	U	Q	R	I	M	I	R	L
T	P	E	T	E	O	S	N	L	Q	E	E	L	M	I	R	I	N	E	A
I	J	E	H	I	R	V	E	E	U	A	W	A	C	G	M	D	T	S	T
O	C	X	P	N	W	E	S	L	I	S	L	R	H	H	S	S	E	P	I

PARALLELLINES

PERPENDICULAR

TRANSVERSAL

ANGLERELATIONSHIP

CORRESPONDING

ALTERNATEINTERIOR

ALTERNATEEXTERIOR

CONSECUTIVEINTERIOR

RIGHTANGLE

LINESEGMENTS



Republic of the Philippines
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National Capital Region
SCHOOLS DIVISION OFFICE OF QUEZON CITY
MALIGAYA HIGH SCHOOL
Ilang Ilang St., Maligaya Park Subd.
Pasong Putik, Nova, Quezon City

Name: _____
Section: _____

Date: _____
Score: _____

Third Summative Test in Mathematics 8
S.Y. 2023–2024

Multiple Choice: *Choose the letter that corresponds to the correct answer. Write the answer in your answer sheet.*

1. A point can be represented as:
A. Blackboard B. Ruler C. Scissors D. Tip of a pen
2. Which of the following best represents a plane?
A. Blackboard B. Ruler C. Scissors D. Tip of a pen
3. A structure that consists of defined and undefined terms, axioms or postulates, and theorems is called:
A. Direct proof B. Indirect proof C. Law of Syllogism D. Mathematical system
4. Two lines that are not parallel ____ intersect.
A. always B. sometimes C. maybe D. never
5. Evaluate the following statement: "Perpendicular lines form right angles."

Grade 8 Fourth Quarter Reviewer

Jonathan R. Bacolod, LPT

TeXMathPro: Math Teaching Beautified

March 10, 2025

Applying Theorems on Triangle Inequality

The Triangle Inequality Theorem states that the sum of the lengths of any two sides of a triangle must be greater than the length of the third side:

$$a + b > c, \quad a + c > b, \quad b + c > a$$

Basic Concepts of Probability

Probability measures the likelihood of an event occurring, expressed as:

$$P(A) = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}}$$

where $0 \leq P(A) \leq 1$.

Counting the Outcomes of Experiments

- Use the Fundamental Counting Principle: If one event has m outcomes and another event has n outcomes, the total number of outcomes is $m \times n$.
- Tree diagrams and tables help visualize possible outcomes.

Experimental and Theoretical Probability

- **Theoretical Probability:** Based on mathematical reasoning.
- **Experimental Probability:** Based on actual experiments.

Parallelism and Perpendicularity

- Parallel lines never intersect and have the same slope.
- Perpendicular lines intersect at 90° and their slopes are negative reciprocals.

Probability of an Event

Given an event A , its probability is calculated as:

$$P(A) = \frac{\text{Favorable Outcomes}}{\text{Total Possible Outcomes}}$$

If A' is the complement of A :

$$P(A') = 1 - P(A)$$

Proving Inequalities in a Triangle

To prove triangle inequalities, use:

- Triangle Inequality Theorem.
- Exterior Angle Inequality Theorem.
- Hinge Theorem.

Proving Properties of Parallel Lines Cut by a Transversal

When two parallel lines are cut by a transversal:

- Corresponding angles are congruent.

$$4\sqrt{6} + 3\sqrt{6}$$

242

$$6\sqrt{5}$$

TeXUser[®]Pro: Math Teaching Assistant
 Created by Jonathan R. Borland, LPT
 Adding Radical Expressions

$$5\sqrt{7} + \sqrt{7}$$

 $\triangle ABE + \triangle ADE$

$$3\sqrt{10} + 4\sqrt{10}$$

484

$$9\sqrt{2}$$

Adding Radical Expressions

$$\sqrt{3} + 4\sqrt{3}$$

EAS

$$4\sqrt{6} + \sqrt{6}$$

3/5

$$3\sqrt{2}$$

TyXMod[®]Proc Math Teaching Resources
 Created by Jonathan R. Dacout, LPY
 Adding Radical Expressions

$$4\sqrt{7} + 6\sqrt{7}$$

$$6\sqrt{5} + 2\sqrt{5}$$

72 Axiom

5

Adding Radical Expressions

$$\sqrt{8} + 3\sqrt{2}$$

2006.10.14

$$2\sqrt{3} + \sqrt{12}$$

$$7\sqrt{2}$$

$$2 + \sqrt{4}$$

Factoring Trinomials Flashcards

by Jonathan R. Bacolod, LPT

Flashcard 1 of 10

$$n^2 - 2np + p^2$$

Type factored form, e.g. $(2x + 1)(x - 9)$

Previous

Next

Submit

Record Sheet in Math

Leader's Name:

Jonathan R. Bacolod

Section:

8-Malaysia

Group Number:

3

Last Name	First Name	3.1	3.2								
Bacs	Jon	7	5								

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calibratable bubble grid

Live — start Auto-Scan.

Setup

The 3 things you set per test.

Section

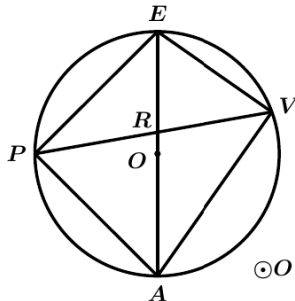
7-Malvar

Questions

50

Practice Exercises

A. Refer to the figure on the right to answer the following.



1. Name the angles that intercept $\widehat{AP}$.
2. Name the angles that intercept $\widehat{EV}$.
3. Name the arc that is intercepted by $\angle PAE$.
4. Name the arc that is intercepted by $\angle EVP$.
5. If $m\angle PEA = 48$, then $m\widehat{AP} = \underline{\hspace{1cm}}$ and $m\angle AVP = \underline{\hspace{1cm}}$.
6. $m\angle EPA = \underline{\hspace{1cm}}$
7. $m\angle EVP + m\angle PVA = \underline{\hspace{1cm}}$
8. If $m\angle VEP = 100$, then $m\angle PAV = \underline{\hspace{1cm}}$.



Jonathan Rufo Bacolod > All Books

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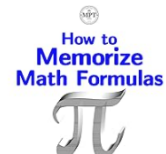


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How to Memorize Math Formulas

A Comprehensive Guide to Memorization and Retention for High School Teachers and Students

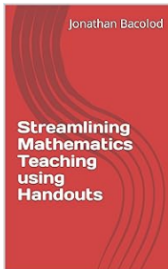
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Quick look

How to Memorize Math Formulas: A Comprehensive Guide to Memorization...

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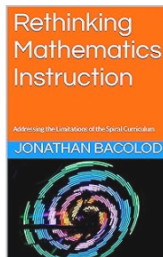
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Quick look

Streamlining Mathematics Teaching using Handouts

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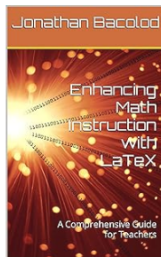
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Quick look

Rethinking Mathematics Instruction: Addressing the Limitations of the...

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Quick look

Enhancing Math Instruction with LaTeX: A Comprehensive Guide for Teachers

Kindle Edition\$0.00 **kindleunlimited**

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WHAT APP?

AI CHATBOT

L

L

M

LARGE LANGUAGE MODELS

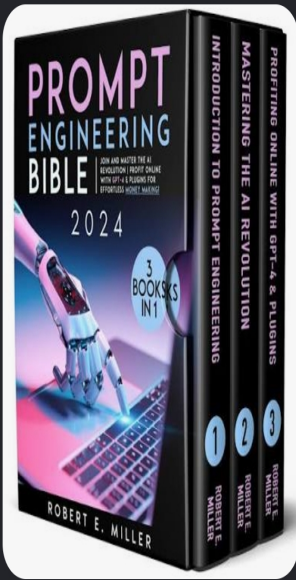
СНАТ

Why can't you harness the power of
AI?

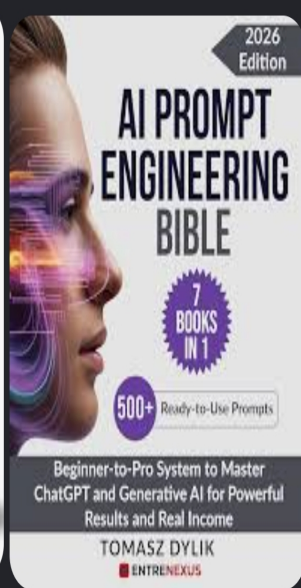
**Use AI to Create, Adapt, and
Improve Learning Materials**

The SCARE Framework: Creating Effective AI Prompts for Teaching and Learning

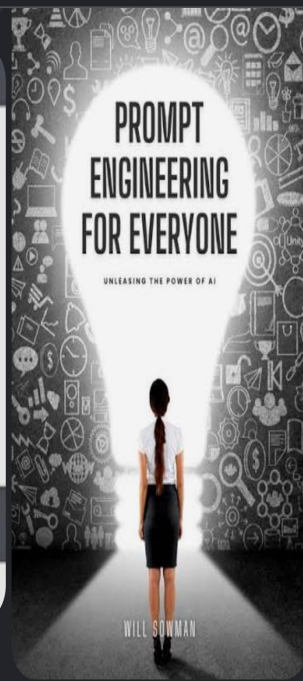
Jonathan R. Bacolod



Prompt Engineering Bibl...



AI Prompt Engineering...



AI Prompt Engineering A...
Shoppable - Shoppable...



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Transforming Math Education



Empowering High School Teachers with AI and Prompt Engineering

Jonathan R. Bacolod, LPT

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Transforming Math Education: Empowering High School Teachers with AI and Prompt Engineering

by [Jonathan Bacolod](#) (Author) | Format: Kindle Edition

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Transforming Math Education: Empowering High School Teachers with AI and Prompt Engineering is a comprehensive guide that explores the integration of artificial intelligence (AI) and prompt engineering techniques in high school math education. The book covers various foundational aspects of prompt engineering, such as understanding AI language models, principles of prompt engineering, and writing

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Workshop Outcome

Every teacher will create **3–5 AI-assisted teaching materials** that are relevant to their own subject, learners, and classroom context.

WHAT CAN AI HELP YOU CREATE?

LESSON PLANS AND ACTIVITIES

WORKSHEETS

HANDOUTS

STUDY NOTES

QUIZZES

ASSESSMENTS

ANSWER KEYS

EXAMPLES

EXPLANATIONS

GUIDED PRACTICE

DIFFERENTIATED ACTIVITIES AND SCAFFOLDS

RUBRICS
CHECKLISTS
FEEDBACK GUIDES

DISCUSSION QUESTIONS REFLECTION PROMPTS

REAL-WORLD APPLICATIONS PERFORMANCE TASKS

Across Departments

The same prompting principles can be used in every subject.

MATHEMATICS

Practice problems and error analysis

SCIENCE

Investigation guide and explanation

ENGLISH / FILIPINO

Writing prompts and feedback guide

AP / SOCIAL STUDIES

Source-analysis questions

TLE / TVL

Procedure checklist and rubric

МАРЕН

Performance-task criteria and reflection

TRY IT



◆ Upgrade



Memory off ⓘ



Create better and faster images ›

Edit your photos, make images of yourself, and try new styles



Ask ChatGPT





Afternoon, Jon

Get more with Claude Pro

[Upgrade to Pro](#)

Chat with Claude...



Sonnet 5 Medium





Flash Extended ▾



What can I help with, ** Jonathan?

+ Ask Gemini



Try It

“Create a worksheet.”

AI CAN PRODUCE SOMETHING.

BUT WHAT EXACTLY SHOULD IT
CREATE?

FOR WHICH SUBJECT?

FOR WHICH LEARNERS?

ABOUT WHAT COMPETENCY?

HOW MANY ITEMS?

WHAT DIFFICULTY?

WHAT FORMAT?

A Better Prompt

Science example:

“Create a 15-item Grade 8 worksheet on separating mixtures. Include 5 identification items, 5 classification items, and 5 short-response items. Use familiar household examples and provide an answer key. Format the output as downloadable docx file.”



PROMPT ENGINEERING

WHAT IS PROMPT ENGINEERING?

DESIGN INSTRUCTIONS

REFINE INSTRUCTIONS

SO AI

UNDERSTANDS

THE TASK

THE CONTEXT

THE CONSTRAINTS

THE DESIRED
OUTPUT

The Basic Idea

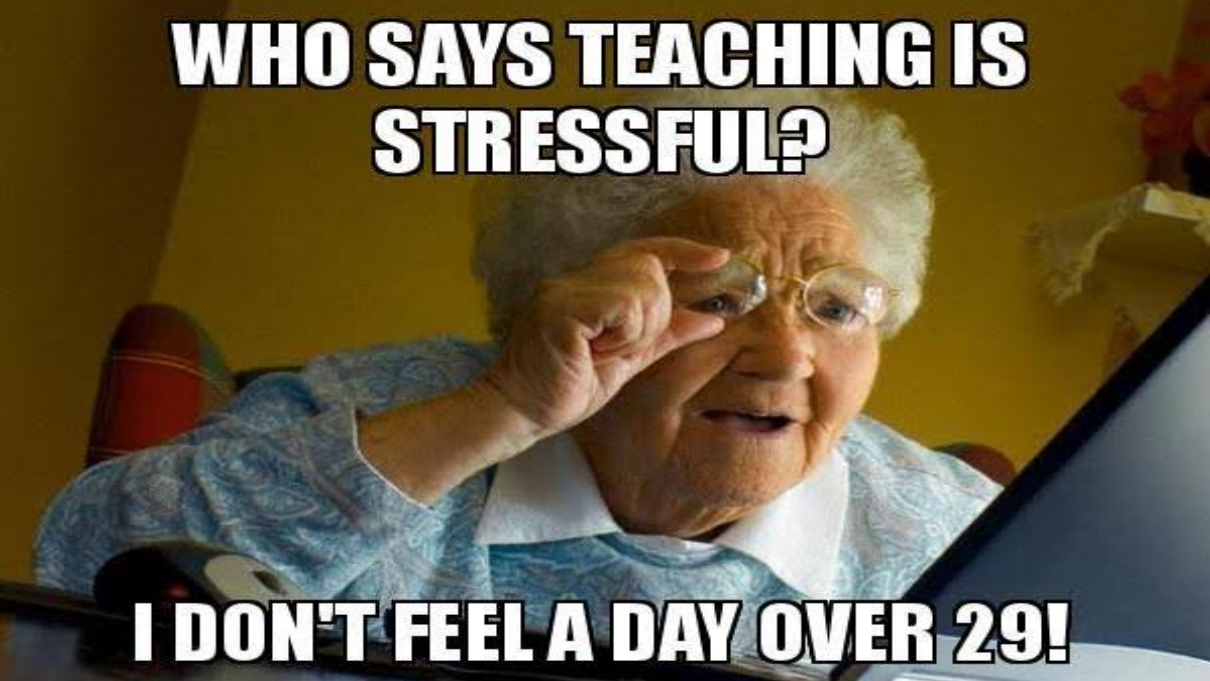
**Better input $\longrightarrow$ better-aligned
output**

But AI output still needs teacher
judgment, checking, and revision.

WHY CARE?

Save time on repetitive drafting.

**WHO SAYS TEACHING IS
STRESSFUL?**



I DON'T FEEL A DAY OVER 29!

Generate multiple versions of an activity.

Adapt materials for different learners.

Brainstorm examples and applications.

Turn one resource into several formats.

Get a starting point when creating something from scratch.

**AI is a drafting and thinking
assistant,
not the final decision-maker.**

THE SCARE FRAMEWORK

S — SPECIFIC

C — CONTEXT

A — ASSIGN A ROLE

R — REFINE

E — EXAMPLES

Use SCARE as a checklist
when a prompt is not producing
the result you need.

PRINCIPLE #1

BE SPECIFIC

A SPECIFIC PROMPT TELLS AI
WHAT TO PRODUCE.

task or product

topic or competency

number of items or sections

learner level

desired format

difficulty or cognitive level

SPECIFICITY
ACROSS SUBJECTS

Mathematics

“Create 10 Grade 9 problems on solving quadratic equations by factoring”

Science

“Create 8 Grade 7 questions on classifying mixtures as homogeneous or heterogeneous.”

English

“Create 10 Grade 8 items that distinguish fact from opinion in short passages.”

TLE

“Create a 12-step checklist for safely preparing a workstation before a practical activity”

Vague vs. Specific

TRY IT

“Make a lesson about climate change.”

“Create a 45-minute Grade 9 lesson on the causes and effects of climate change. Include a 5-minute activating question, a short teacher explanation, a group data-analysis activity, three processing questions, and a 5-item exit ticket.”



BEFORE SENDING THE PROMPT...

Did I state the exact task?

Did I identify the topic or competency?

Did I specify the learner level?

Did I specify quantity?

Did I specify the output format?

Did I state important constraints?

Her : why can't you be more pacific ??
Me : say no more...



PRINCIPLE #2

GIVE CONTEXT

USEFUL CONTEXT MAY INCLUDE...

grade level or age

prior knowledge

misconceptions or difficulties

reading or language level

interests and experiences

available technology and materials

class size and time available

Context Changes the Output

TRY IT

Compare:

“Explain photosynthesis.”

“Explain photosynthesis to Grade 7 learners who can identify plant parts but struggle to connect sunlight, carbon dioxide, water, and glucose. Use simple language, one analogy, and a short concept-check.”



YOU CAN TELL AI...

“My students are struggling with...”

“They already know...”

“Their reading level is approximately...”

“I have only 15 minutes.”

“There is no internet or student device.”

“Use examples familiar to Filipino students.”

“The activity must work for 40
learners.”

ALSO PROVIDE CONTEXT
ABOUT THE TEACHING
SITUATION

curriculum competency or learning
objective

assessment purpose

available resources

time limit

classroom constraints

required school format

PRINCIPLE #3

ASSIGN A ROLE

A ROLE CAN FOCUS
THE KIND OF RESPONSE YOU
WANT.

curriculum planner

lesson designer

assessment designer

differentiation specialist

Socratic tutor

writing coach

activity designer

rubric designer

feedback assistant

subject-matter explainer

Roles Across Departments

Science: “Act as an experienced science teacher who designs inquiry-based investigations.”

English: “Act as a writing coach who gives constructive feedback without rewriting the student’s work.”

TLE: “Act as a practical-skills instructor who prioritizes safety and clear procedures.”

Mathematics: “Act as an assessment designer who checks mathematical accuracy and alignment with the competency.”

Role + Task

Instead of:

“Make a quiz.”

Role + Task

try:

“Act as an assessment designer.
Create a 15-item quiz aligned with
the following competency...”

**ONLY WAY TO KEEP
AI FROM TAKING YOUR JOB**

**IS TO USE AI TO
DO YOUR JOB BETTER**

PRINCIPLE #4

REFINE

The First Output Is a Draft

First response $\neq$ Final response

Prompting is iterative.

YOU CAN REFINE...

difficulty

number of items

format

vocabulary and reading level

length

cognitive level

constraints

answer format

examples and contexts

visuals or tables

hints and scaffolds

tone and language

Examples of Refinement

“Make the questions easier.”

Examples of Refinement

“Keep the same competency, but reduce the reading load, use familiar contexts, and make the calculations require only one major step.”

Examples of Refinement

“Keep the content, but convert the questions into multiple choice with four plausible options and an answer key.”

REFINE THE FORMAT

“Put the answer key in a separate section.”

“Use a table with three columns.”

“Turn this into a one-page handout.”

“Use numbered steps.”

“Format the mathematics for LaTeX.”

“Use language suitable for Grade 7 learners.”

ADD LEARNING SUPPORT

worked examples

hints

guiding questions

sentence starters

vocabulary support

partially completed examples

extension questions

remediation items

PRINCIPLE #5

SHOW EXAMPLES

SHOW AI WHAT YOU WANT

a sample question

a sample answer

a preferred format

a preferred tone

a table or template

an example of what to avoid

Show the Style

If you want a particular style, provide an example.

Show the Style

“Write the new word problems in this style: a short classroom or community situation, one clear question, realistic quantities, and no unnecessary information.”

Examples reduce ambiguity.

Show What to Avoid

You can also provide a negative example.

Show What to Avoid

“Do not use advanced vocabulary, trick questions, culturally unfamiliar situations, or information that is not needed to solve the problem.”

PUT SCARE TOGETHER

A Complete Prompt

Specific: Create a 10-item assessment on ecosystems. **Context:** Grade 8 learners have studied food chains but struggle with food webs. **Role:** Act as an experienced Grade 8 science assessment designer. **Refine:** Use four-choice multiple choice, moderate difficulty, and familiar local examples. **Examples:** Follow this item format: question, choices A–D, correct answer, one-sentence rationale.

SCARE Is a Checklist, Not a Formula

A short prompt can still be strong if it contains the information AI needs.

PROMPT MAKEOVER

“Make a Lesson About Fractions”

“Act as a Grade 7 mathematics lesson designer. Create a 45-minute lesson on adding fractions with unlike denominators. My learners know equivalent fractions but often forget to find a common denominator. Include an activating question, worked example, guided practice, pair activity, and 5-item exit ticket. Use clear Grade 7 language.”

“Make Difficult Problems.”

“Create 8 Grade 9 mathematics problems on quadratic functions at an analyzing level. Include a mixture of tables, equations, graphs described in words, and short real-world contexts.

Require students to compare representations and explain relationships. Provide an answer key but not full solutions.”

“Explain Photosynthesis.”

“Act as a Grade 7 science teacher. Explain photosynthesis to learners who know the basic parts of a plant but struggle to connect inputs and outputs. Use simple vocabulary, one analogy, a word equation, and three concept-check questions. Keep the explanation under 250 words.”

“Make a Quiz.”

“Act as an assessment designer.
Create a 15-item Grade 8 English quiz
on distinguishing fact from opinion.
Use short, age-appropriate statements.
Include 8 identifying items and 7 items
requiring students to justify their
choice. Provide A–D choices and a
separate answer key.”

“Create Slides.”

“Create an 8-slide lesson sequence for Grade 10 TLE on basic workplace safety. Each slide should have one main idea, no more than five bullets, one practical example, and one question for students. End with a short safety checklist.”

USE AI RESPONSIBLY

THIS HERE IS CALLED AI

**YOU ASK IT QUESTIONS,
AND IT CONFIDENTLY LIES TO YOU**

AI CAN PRODUCE...

incorrect facts

incorrect calculations

weak examples

invented or unreliable references

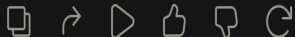
ambiguous questions

inappropriate difficulty

biased or unsuitable wording



The image is just cropped a little differently, so it's still the inverted white-on-black version and would still trip up most scanner apps. If you'd like, I can generate that worksheet now.



Claude is AI and can make mistakes.

Reply to Claude...



Sonnet 5 Medium





Everything should be made as simple as possible, but not simpler.



Gemini is AI and can make mistakes.

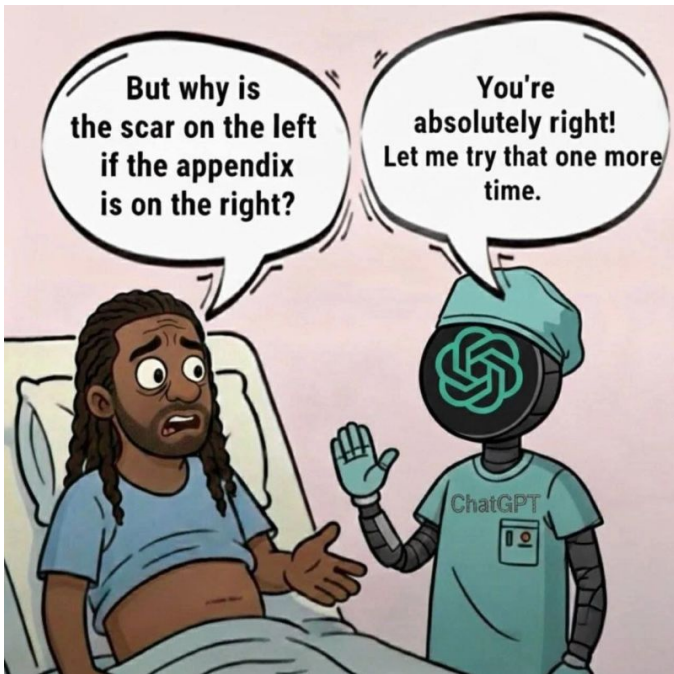


Ask Gemini



But why is
the scar on the left
if the appendix
is on the right?

You're
absolutely right!
Let me try that one more
time.



A confident answer
is not automatically
a correct answer.

THE TEACHER REMAINS
RESPONSIBLE
FOR DECIDING WHETHER THE
OUTPUT IS...

accurate

aligned with the competency

appropriate for the learners

culturally and contextually suitable

instructionally useful

safe and responsible to use

BEFORE GIVING AI-GENERATED
MATERIAL
TO STUDENTS, CHECK...

Content accuracy

Alignment with the learning objective

Level of difficulty

Language and clarity

Answer-key accuracy

Bias, stereotypes, and inappropriate
content

Names, dates, statistics, and references
when relevant

A two-panel comic strip. The top panel shows a man with dark hair and a yellow shirt looking skeptical. The bottom panel shows a man with a mustache and a grey shirt pointing his finger. The background of both panels is an office with a staircase and bookshelves.

Researchers

How do we know that you're
not fabricating all this?

ChatGPT 

That's the neat part. You don't.

PRACTICE

CHOOSE ONE LESSON

subject

grade level

competency or objective

time available

learner difficulty or need

WRITE THE VAGUE PROMPT

“Make a worksheet about...”

“Explain...”

“Create a quiz on...”

“Make an activity for...”

Do not try to make it perfect yet.

ADD SCARE

Specific task

Context about learners and classroom

Assigned role, when useful

Refinements and constraints

Examples or desired format

GENERATE
THEN REVIEW

What worked?

What is inaccurate or unclear?

What does not fit my learners?

What should be changed?

REFINE AGAIN

“Keep the questions, but lower the reading level.”

“Replace items 3 and 7.”

“Increase the cognitive demand.”

“Add an answer key with brief rationales.”

“Make this fit on one page.”

Workshop Challenge

Choose one lesson
from your own subject.

Using SCARE,
create at least **three**
useful materials.

WORKSHEET

ASSESSMENT

ANSWER KEY

SLIDES

HANDOUT

STUDY NOTES

FLASHCARDS

RUBRIC

ACTIVITY GUIDE

REVIEW MATERIAL

SCARE

1. SPECIFIC

Say exactly what you want.

2. CONTEXT

Tell AI who, what, and where.

3. ASSIGN A ROLE

Give AI a useful perspective.

4. REFINE

Improve the output through iteration.

5. EXAMPLES

Show the format or style you want.

Don't be SCAREd of AI.

Use SCARE to make AI useful to **your
subject, your learners, and your
classroom.**

QUESTIONS?